



DEALER BEST PRACTICE SERIES

Utilizing exchange program for slow-turn salvaged parts to improve CRC TAT

Application Maintenance Site Management Component Rebuild Safety MARC Management

Improving CRC Turnaround through Machine Shop Subcomponent Exchange Program . **Error! Bookmark not defined.**

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1.0 Introduction

During the process of rebuilding major components, select parts are often sent to a machine shop for salvage work. The machine shop may be dealer owned, or an outside contractor. Lead-time for the salvage work to be completed causes a delay in the rebuild process. Delays and interruptions to the rebuild workflow further cause inefficiencies, which impact operational performance and delivery.

The following inefficiencies are resultant from excessive delays in the workflow.

- Parts often need to be moved for temporary storage.
- Parts may need to be cleaned again before assembly.
- Parts can get lost or disorganized.
- The work in process time is increased thus impacting shop capacity.

An effective solution is to create an “exchange” inventory of commonly salvaged subcomponents, readily available to eliminate machine shop salvage delays.

2.0 Best Practice Description

One dealer conducted an analysis of turnaround time for subcomponents, which were sent to the machine shop for salvage. Research found that some subcomponents typically experienced a 10-day lead-time from the machine shop back to the rebuild center. The delays were causing bottlenecks, subsequently affecting shop efficiency.

The decision was made to create a “subcomponent exchange” program for those parts identified as causing significant delays to the rebuild process.

Criterion established:

If Salvage Lead-time:

1. Greater than 5 days, and
2. Turns more than 10 times annually.

Then:

**** Stock spare subcomponent inventory at the Machine Shop**

Subcomponents, which fell into this scenario typically, included:

- Engine blocks
- Crankshafts
- Spindles.

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Improving CRC Turnaround through Machine Shop
Subcomponent Exchange Program

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3.0 Implementation Steps

The “machine shop subcomponent exchange” concept makes a good 6 Sigma project. Implementation steps would include:

- **Define** the impact on shop efficiency as result of delays in waiting for machine shop salvage.
- **Measure** the delays and identify which subcomponents cause the delays.
 - Measure the impact in terms of less labor efficiency and impact on work in process.
 - If you have a major component exchange inventory, then improving component turnaround time may allow you to reduce your component inventory level.
- **Analyze** cost versus benefit of stocking exchange subcomponents and payback time/volume.
- **Improve.** Create subcomponent exchange program and document the benefits.
- **Control.** Monitor impact on labor efficiency and shop turnaround time.

4.0 Benefits

- Labor efficiency due to uninterrupted workflow, less material handling, less re-cleaning of parts.
- Increased shop capacity due to better component turnaround time.
- Future reduction to major component exchange inventories due to better turnaround time.
- Less shop space utilized for temporary storage.
- Better overall use of assets.

5.0 Resources Required

Investment in exchange subcomponents.

6.0 Supporting Attachments / References

None applicable at this time.

7.0 Related Best Practices

0906-4.5-1013 - *Improving CRC Component Turnaround*

8.0 Acknowledgements

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